

TEPL 1420 – Plumbing Apprentice IVB (3 Credits)

Course Description

The Plumbing IVB course assists the apprentices in passing the state test and provides a foundation for success as a journeyman plumber. A variety of modules are available for the student to work from, with a focus on what each individual student needs in order to prepare for the test.

Course Objectives

- Review International Plumbing Code (IPC) chapters.
- Apply mathematics for plumbers and pipefitters.
- Use International Fuel Gas Code for gas pipe installation and sizing.
- Define traps and commercial applications.
- Explain storm drainage and commercial/industrial applications.
- Identify boilers, water heaters, and pressure vessels.
- Perform applied trade formulas using different equations.
- Calculate heat loss vs. radiator size.
- Determine radiation sizing for total heat loss of a room.
- Use Journeyman skills such as takeoff lists, leadership skills, communication, and basic business skills.
- Fabricate several piping projects in a lab situation.

Course Outline

- Year IVB

Textbook & Reading Materials

International Fuel Gas Code 2021 (ISBN: 9781609839666)

International Fuel Gas Code 2021 Study Companion (ISBN: 9781955052320)

International Plumbing Code 2021 (ISBN: 9781609839628)

International Plumbing Code 2021 Study Companion (ISBN: 9781955052283)

Assignments & Assessments

Week 1

Hybrid Assignments

Week 2

Work Information

Math Assignment: How Well Do I Know My Stuff?

Requirements for the Plumber Journeyman Test

Math: Hydraulics Review

48 Math Review

Hybrid Assignments

Flash Cards-2

Math Review Quiz

Read IFGC Chapter 4: Gas Piping Installations Sections

401 - 406 including the Tables

IFGC (401-406) SS 4

Week 3

Hybrid Assignment

Week 4

IFGC Pipe Sizing Worksheet 1

IFGC Pipe Sizing Worksheet 2

Read IFGC Chapter 4: Gas Piping Installations Sections

407 - 416

IFGC Ch. 4 (407-416) SS 5

Hybrid Assignments

Week 5

Read Chapter 10 IMC Boilers, Water Heaters and Pressure Vessels in the IMC code book

IMC Chapter 10 Study Session 11

Read Chapter 12: Hydronic Piping in the IMC Code Book

IMC Chapter 12 Study Session 13

Hybrid Assignments

Week 6

Math: Rolling Offsets

Math: PVC Labs #1-5

Math: PVC Labs #6-10

Hybrid Assignments

Read Modern Plumbing Chapter 29

Watch

Week 7

NFPA 13D Chapters 1 to 3

NFPA 13D Chapters 4 to 6

NFPA 130 Chapter 7

NFPA 13D Chapter 8 Part 1

NFPA 13D Chapter 8 Part 2

NFPA 13D Chapter 9

NFPA 130 Chapters 10 to 12

Hybrid Assignments

Week 8

Review IPC Chapter 7: Sanitary Drainage

dd IPC ss 9 Quiz - Sanitary Drainage I

dd IPC 55 10 Quiz - Sanitary Drainage II

Review IPC Chapter 8: Indirect/Special waste

dd IPC SS 11 Quiz - Indirect/Special Waste

Hybrid Assignments

Review Chapter 9 Vents in I PC

dd IPC SS 12 Quiz - Vents I

dd IPC 5513 Quiz - Vents II

Review IPC Chapter 11: Storm Drainage and Chapter 12: Special Piping and Storage Systems

dd IPC SS 15 Quiz - Storm Drainage, Special Piping and Storage Systems, Nonpotable Water Systems and Subsurface Landscape Irrigation Systems

Study Resources

IPC 11 Roof Drain Sizing.pptx

Week 9

Hybrid Assignments

Review IPC Chapter 6: Water Supply and Distribution

dd IPC SS 7 Quiz - Water Supply and Distribution I

dd IPC SS 8 Quiz - Water Supply and Distribution II

Review IPC Chapter 10: Traps, Interceptors, and Separators

dd IPC SS 14 Quiz - Traps, Interceptor and Separators

Week 10

Hybrid Assignments - Make up Days

Week 11

Hybrid Assignments

Review IPC Chapter 3: General Regulations

Week 12

dd IPC SS 2 Quiz - General Regulations I

dd IPC SS 3 Quiz - General Regulations II

Review IPC Chapter 4: Fixtures, Faucets and Fixture Fittings

dd IPC 55 4 Quiz - Fixtures, Faucets and Fixture Fittings I

dd IPL SS 5 Quiz - fixtures, faucets and fixture fittings II

Hybrid Assignments

Review IPC Chapter 5: Water Heaters

IPC SS 6 • Water Heaters

On Campus Skills Pass-off

Week 13

Lab Activity: PVC Project # 3 Using 1 and 3/4 PVC Pipe and Fittings

Lab Activity: PVC Project #9 Rolling Offset Using 1 and 3/4 PVC Pipe and Fittings

Lab Activity: PVC Project #2 Using 1 and 3/4 PVC Pipe and Fittings

Hybrid Assignments
Final Competency Exam Review
Final Exam Review
Week 14
On Campus Skills Pass-off

Final Competency Exam
Week 15
On Campus Skills Pass-off
Competency Exam Retake

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Start: 01-05-2027

End: 04-21-2027

Tu, W

6:00 PM - 9:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jason Parsons — jparsons@stech.edu

Wade Esplin — wesplin@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

You will also be required to complete the course, exams and homework, with 75% or higher accuracy for credit and DOPL licensure points.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation, and student services will be notified.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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